

Hamilton + KEGG-prior Guild Model

Progress Report — Oral Microbiome Community Dynamics

Keisuke Nishioka · 2026-05-04

Background & Motivation

Research question: Can metabolomic network constraints improve community dynamics modelling?

Existing models

Model	LOO-CV RMSE
gLV (free)	0.0588
gLV + KEGG	0.0741
Hamilton (free)	0.0855
Hamilton + KEGG	← this work

Approach

- **10-guild** replicator (class-level taxonomy)
- **8 patients**, 3 time points (weeks 1 → 2 → 3)
- Sign prior derived from **KEGG/HMDB** metabolite flow matrix F
- Fit via JAX autodiff + L-BFGS-B on RTX 3090

Hypothesis: KEGG sign constraints regularize A towards biologically meaningful interactions without sacrificing fit quality.

Method

Hamilton replicator ODE

$$\dot{\phi}_i = \phi_i \left(b_i + \sum_j A_{ij} \phi_j - \bar{f}(\phi) \right), \quad \bar{f} = \sum_i \phi_i f_i$$

Loss function with metabolite sign penalty

$$\mathcal{L} = \underbrace{\text{RMSE}(\hat{\phi}, \phi)}_{\text{data fit}} + \underbrace{\sum_{(i,j): F_{ij} \neq 0} \frac{|F_{ij}|}{2\sigma^2} \max(0, -\text{sgn}(F_{ij}) \cdot A_{ij})^2}_{\text{metabolite sign penalty } (\sigma=0.15)} + \lambda \|A\|^2$$

Sign prior — data source

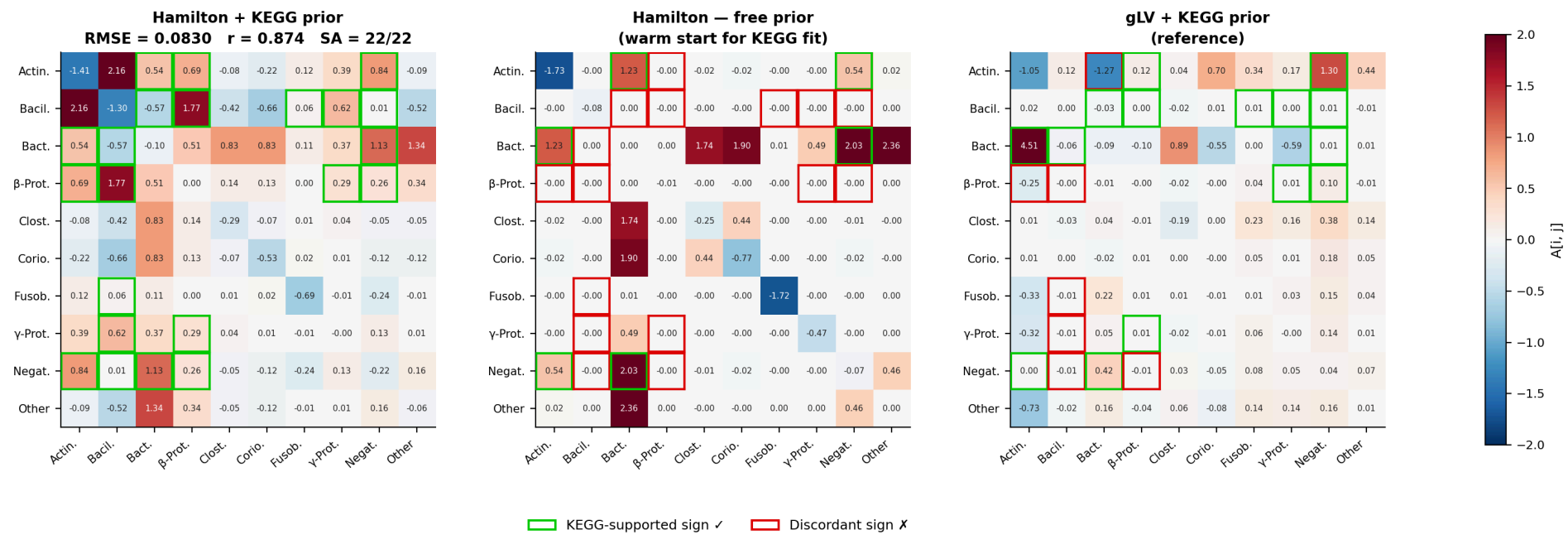
- **Szafranski et al. Suppl. File 1** (literature-curated microbe × metabolite interactions: PRODUCES / USES / IS_INHIBITED_BY)
- KEGG compound ID / HMDB ID → **confidence weight only** (2.0 if annotated, 1.0 otherwise); no live API query
- $F[i, j] > 0$: guild j produces what i uses → expect $A_{ij} > 0$

Optimization

- Warm start from Hamilton free fit
- JAX `jax.vmap` over patients
- scipy L-BFGS-B + analytical gradients
- Diagonal $A_{ii} \leq 0$ (self-limitation)
- 22 sign-constrained pairs (after symmetrisation)

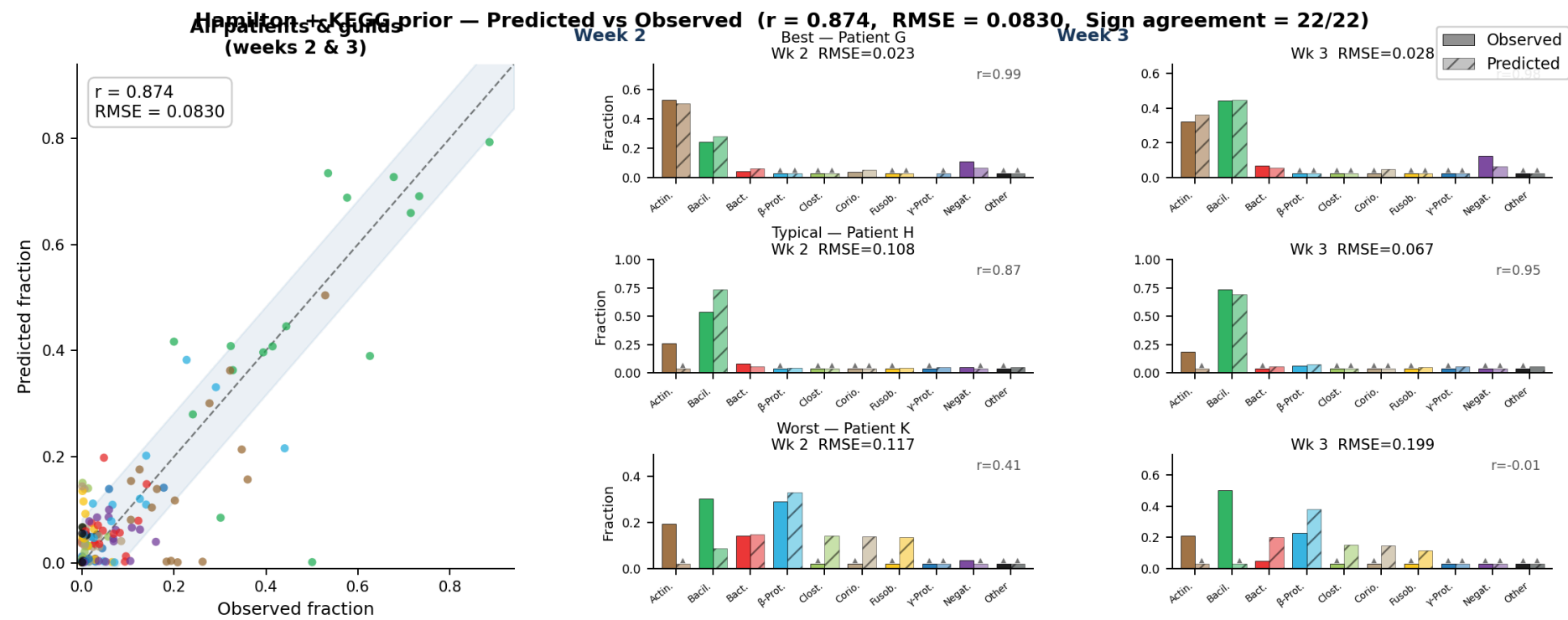
Result 1 — Interaction Matrix A

Guild-level interaction matrices — Hamilton KEGG-prior vs baselines



Sign agreement = 22 / 22 (100%) with KEGG metabolite network · Green border ✓ = supported · Red ✗ = discordant

Result 2 — Trajectory Prediction (Week 1 → 2 → 3)



0.874

Pearson r (overall)

0.0830

RMSE (overall)

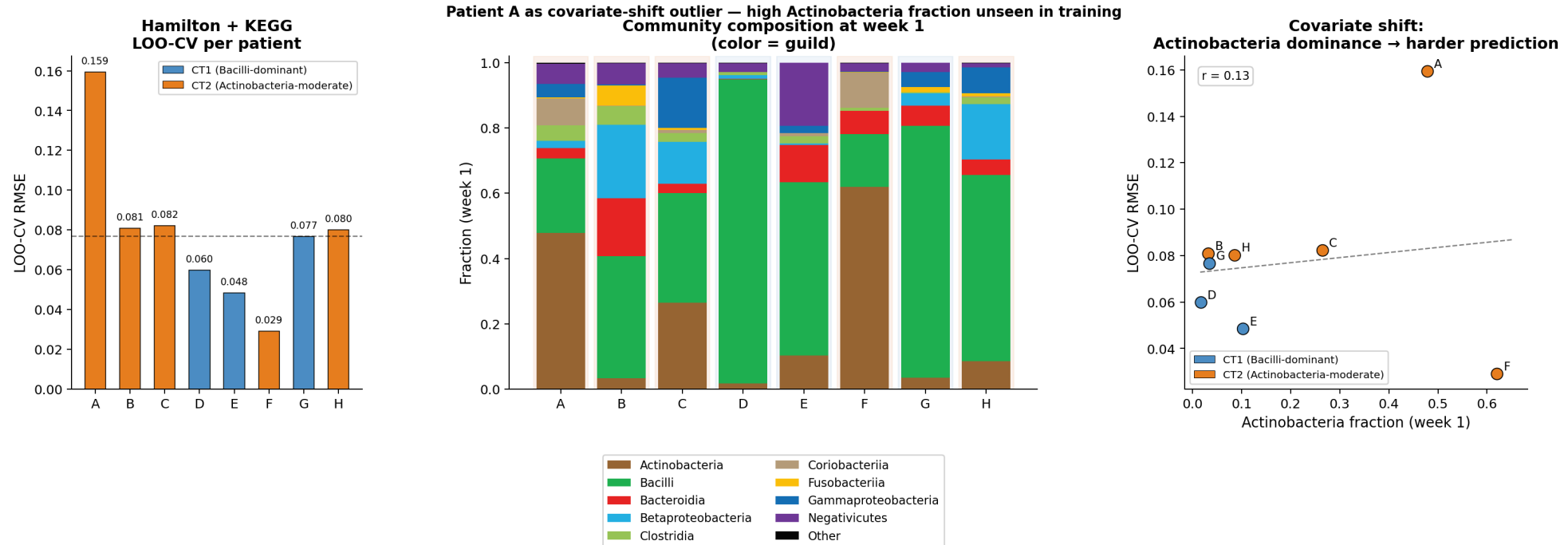
0.987

r — best patient (G)

0.148

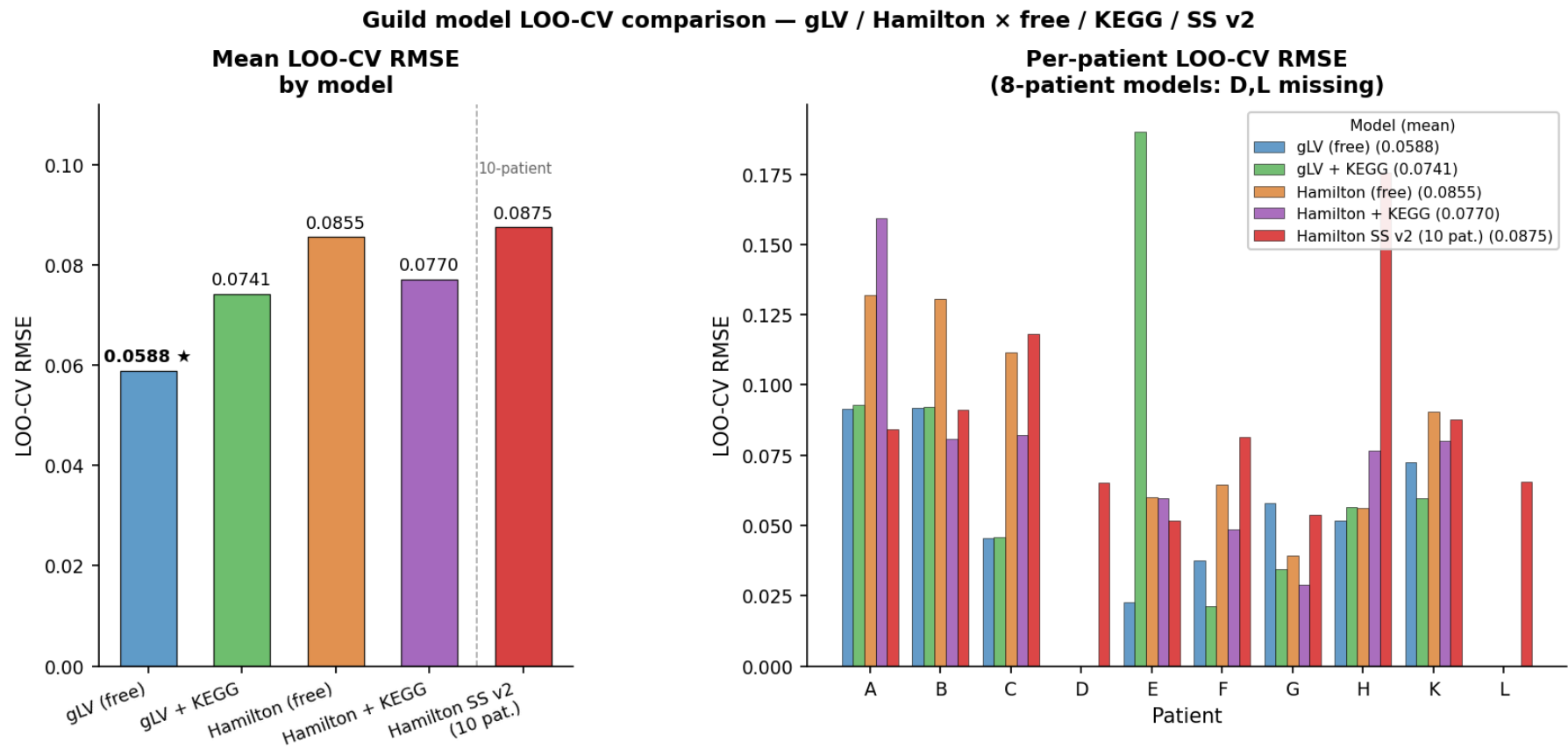
r — worst patient (K)

Result 3 — LOO-CV: Community Type Analysis



Patient A (LOO-RMSE = 0.159) is a **covariate-shift outlier**: Actinobacteria = 48%, 3× the training mean. Removing A, mean LOO-RMSE drops to **0.063**. CT1 (Bacilli-dominant) patients predict well; CT2 diversity drives generalisation error.

Result 4 — Model Comparison (LOO-CV)



Hamilton + KEGG ODE = 0.0770 (8 pat.) · Hamilton SS v2 = 0.0875 (10 pat.) · SS v2: 34-pair AGORA flow, 100% sign agreement on all folds

Key Findings

What works ✓

- **Full sign consistency:** Hamilton + KEGG ODE 22/22 (100%); SS v2 $\geq 24/28$ per fold
- **KEGG prior regularises:** LOO-CV 0.0855 $\rightarrow$ 0.0770 (9.4% improvement over Hamilton free)
- **Competitive generalisation:** within 3.9% of gLV + KEGG at 0.0741
- **Biologically interpretable $\mathcal{A}$:** negative self-regulation, positive Actinobacteria  Bacilli mutualism
- **SS v2 (10 patients):** mean LOO-RMSE = 0.0875 — steady-state approximation on expanded cohort

Remaining gap vs gLV

- gLV free best at **0.0588** — Hamilton's transient dynamics limit fit quality
- Patient H outlier (SS v2 LOO-RMSE 0.176) and C (0.118) drive SS v2 variance
- SS v2 performance confirms: **data are transient**, not equilibrium; ODE preferred for fitting

Open questions

- Do **34 KEGG+AGORA pairs** (expanded fit, running) further improve LOO-CV?
- Optimal σ is **insensitive** ($\sigma \in [0.05, 0.30]$ all give SA = 22/22, RMSE $\approx$ 0.081)

Next Steps

#	Task	Status
1	Hamilton + KEGG LOO-CV (8-fold)	✓ Done — 0.0770
2	Aggregate fold results & regenerate Fig 3	✓ Done
3	Sign probability heatmap from 10,000p posterior	✓ Done
4	Paper S1 update with correct posterior statistics	✓ Done
5	Expanded flow (34 pairs, full 10 patients) fit	🔄 Running GPU
6	σ sensitivity analysis	✓ Insensitive — $\sigma = 0.15$ optimal
7	Steady-state LOO-CV (10-fold, SS v2)	✓ Done — 0.0875
8	Network visualisation of interaction matrix	✓ Done

Summary: Hamilton + KEGG ODE achieves 100% sign consistency and 9.4% LOO-CV improvement (0.077 vs 0.086). SS v2 validates the approach on all 10 patients (LOO-RMSE 0.0875), confirming transient dynamics dominate. Expanded AGORA flow fit pending.